EMPOWER CONFERENCE

# Stop Prompting, Start Building Agents with Criteria

// Building smarter agent architectures in GitHub Copilot + VS Code

Samantha Cruz
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# Speaker



Lucía Muñoz Salas

Power Platform Developer

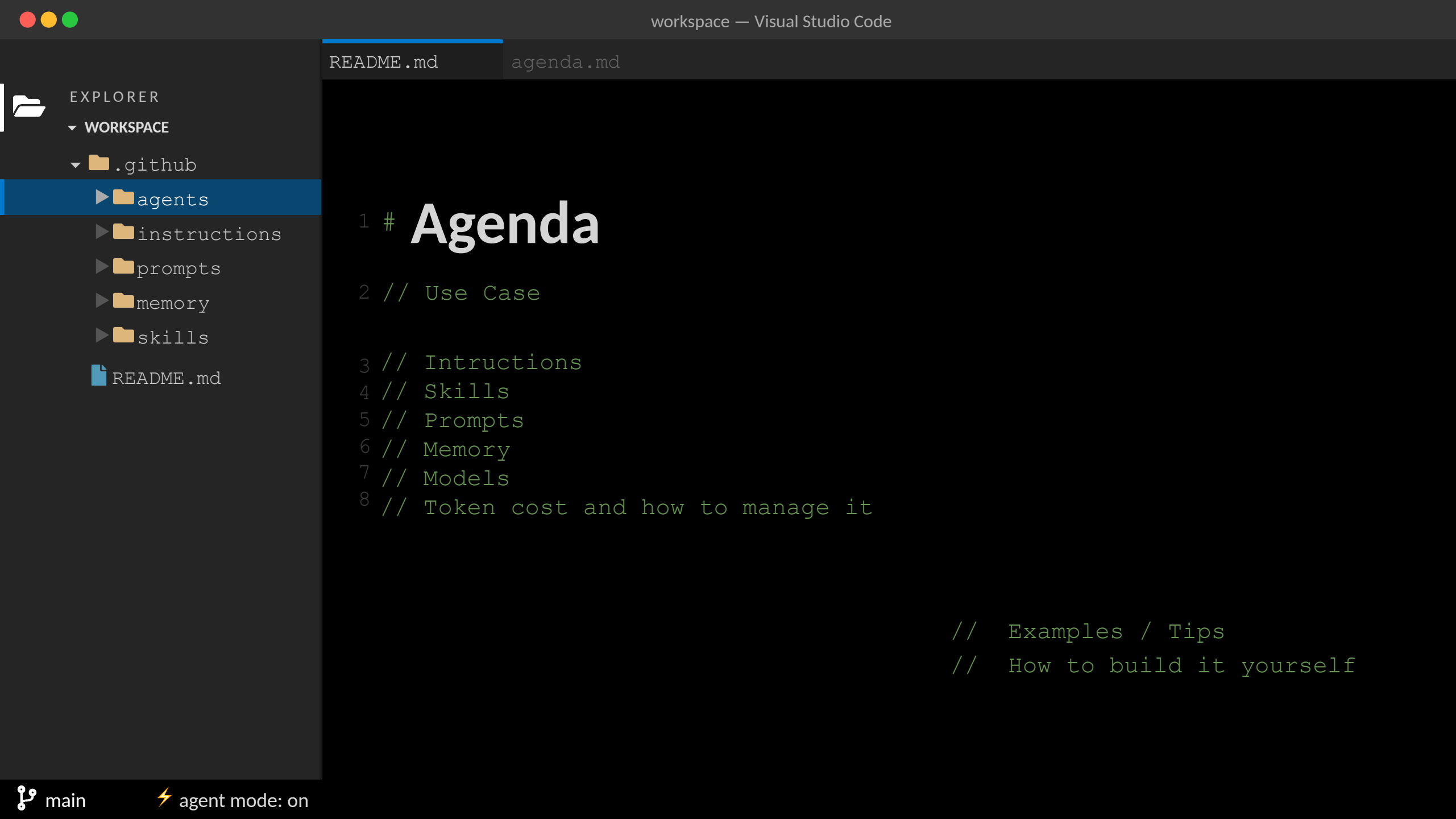
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Samantha Cruz

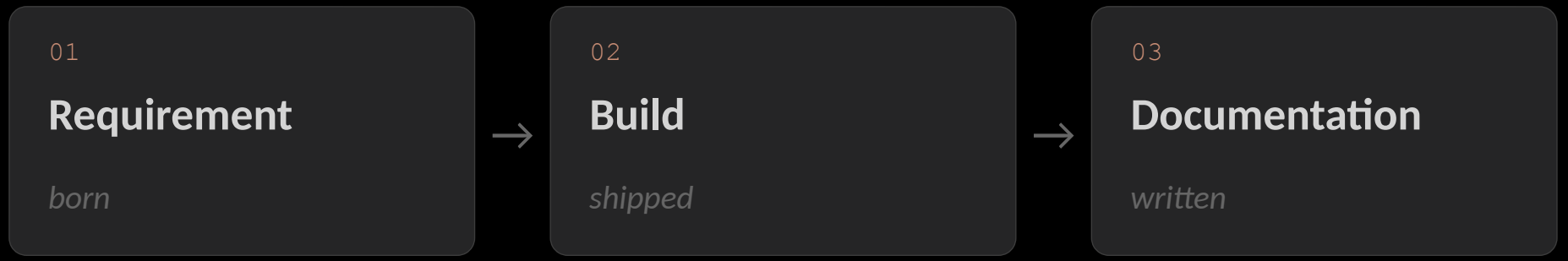
Data Team Lead

isolutions AG


workspace — Visual Studio Code
README.md
agenda.md
EXPLORER
WORKSPACE
.github
agents
instructions
prompts
memory
skills
README.md
# Agenda
// Use Case
// Intructions
// Skills
// Prompts
// Memory
// Models
// Token cost and how to manage it
// Examples / Tips
// How to build it yourself
main
agent mode: on

# AI-First Approach

// "We don't use AI as a tool — we integrate it as part of the team."




// From the moment a requirement is born · to the moment what was done gets documented.

1 // end-to-end process

2 # From customer input to deployed solution.

3 // Four phases, one continuous loop.

01
INPUTS

**Customer story & requirements**

Policies, processes, UX guidelines, acceptance criteria.

02
PLANNING

**Specification agent**

Translates requirements into an actionable design.

03
IMPLEMENTATION

**Specialised build agents**

Data model · Forms & views · Plugins · Flows.

04
RECONCILIATION

**Documentation agent**

Closes the loop — runbooks and as-built reconciliation.

INTEGRATION LAYER

*MCP servers connect agents to the real tools.*

MCP
**Azure DevOps**

MCP
**Dataverse Metadata**

// Specification agent designs · build agents implement · documentation agent reconciles.

```
// what makes it different
```

# Two features that make it robust.

## 01



### Living documentation

```
// Every action writes itself to the wiki.
```

Documentation updates itself and feeds back into the agents as context, so each next decision is better-informed. A self-reinforcing loop.

## 02



### Shared error memory

```
// No mistake is ever repeated twice.
```

When something fails and gets resolved by an agent or a developer's correction, the solution is recorded. Every agent learns. In a distributed team, that's worth a lot.

```
// A self-reinforcing loop · shared learning across the team.
```

# Instructions, Skills and Approvals

## Three ways to teach your agent how you work

### Global Instructions

Your preferences, always on

### Repo Instructions

This project's naming, architecture and conventions

### Skills

Loaded only when the task needs them

## Approvals: how much it can do without you

### Need Approval

Every action waits for you. Slowest, safest

### Bypass Approvals

Actions with low risk skip confirmation

### Auto / Autopilot

Agent approves itself, runs until done or blocked

EXPLORER

AGENT PROJECT

- agents
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  - repo.instructions.md
- memory
- prompts
- skills
  - SKILL.md
  - scripts

01_layers_and_approvals.md

# Instructions, Skills and Approvals

```
---
name: docx
description: "Use this skill whenever the user wants to create, read, edit, or manipulate Word documents (.docx files) or Word templates (.dc
license: Proprietary. LICENSE.txt has complete terms
---

# DOCX creation, editing, and analysis

A `.docx` is a ZIP archive of XML files. Choose your approach by task:

| Task | Approach |
|---|---|
| **Create** a new document | Write a `docx` (npm) script — see gotchas below |
| **Edit** an existing document | `unzip` → edit `word/document.xml` → `zip` (docx-js cannot open existing files) |
| **Read** content | `pandoc -t markdown file.docx` |

> Script paths below are relative to this skill's directory.

## Creating with docx-js — gotchas

`docx` is preinstalled — do not run `npm install` first; write the script and `require('docx')` directly. Only if that require fails: `npm ir

- **Page size defaults to A4.** For US Letter set `page: { size: { width: 12240, height: 15840 } }` (DXA; 1440 = 1″).
- **Landscape:** pass portrait dimensions and `orientation: PageOrientation.LANDSCAPE` — docx-js swaps width/height internally.
- **Tables need dual widths:** set `columnWidths` on the table AND `width` on every cell, both in `WidthType.DXA` (PERCENTAGE breaks in Googl
- **Table shading:** use `ShadingType.CLEAR`, never `SOLID` (renders black).
- **Lists:** never insert `•` literally; use a `numbering` config with `LevelFormat.BULLET`.
- **`ImageRun` requires `type:`** (`"png"`, `"jpg"`, …).
```

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01_layers_and_approvals.md

# Instructions, Skills and Approvals

.github > instructions > global.instructions.md > ## Shared Memory Across Agents

You, 3 months ago | 2 authors (You and one other)

```
---
description: Applies always.
applyTo: '**'
---

## Dataverse Metadata MCP Server

- When updating or creating artifacts using the Dataverse Metadata MCP Server,
  always perform actions sequentially, not in parallel. Parallel execution will fail.

## Azure DevOps

The Azure DevOps Project to use is: `P-AFI-AquaPlus`
Never ask for it, just use it.

When creating work items like tasks or user stories, choose `wit_add_child_work_items` to maintain the parent-child relationship. Only if an item needs to be created independently, use `wit_create_work_items`.

## Known Issues & Solutions

- **Plugin assembly strong-signing required** — Plugin Registration Tool fails with "Assemblies containing Plugins must be strongly signed" if `<SignAssembly>false</SignAssembly>` in .csproj. Fix: generate key with `sn.exe -k <name>.snk`, set `<SignAssembly>true</SignAssembly>` and `<AssemblyOriginatorKeyFile><name>.snk</AssemblyOriginatorKeyFile>`, rebuild. Also: the "Delete AsyncOperation if StatusCode = Successful" checkbox must be unchecked for Synchronous plugin steps — it is only valid for Asynchronous steps.
- **NSG rules do not support FQDN destinations** — NSGs only support IP prefixes, service tags, and ASGs as destinations. Never model NSG rules with FQDN targets. For Azure PaaS services accessed from private subnets, account for Private DNS if Private Endpoints are used, and separate API ingress decisions from backend PaaS connectivity.

## Shared Memory Across Agents

The file `.github/memory/shared--lessons-learned.md` contains a log of non-obvious errors and solutions discovered during development. It has a strict maximum of 20 entries.
```

# Don't Just Use a Public Skill, Make it Yours



## Fork it, don't copy it

A public skill is a starting point, not the finished product



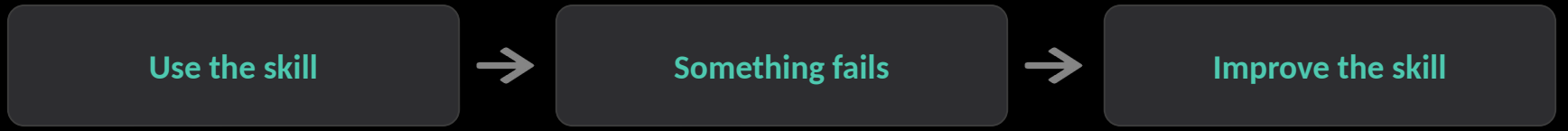
## Inject real project context

Your repo's naming, your layers, your conventions. Not generic best practices

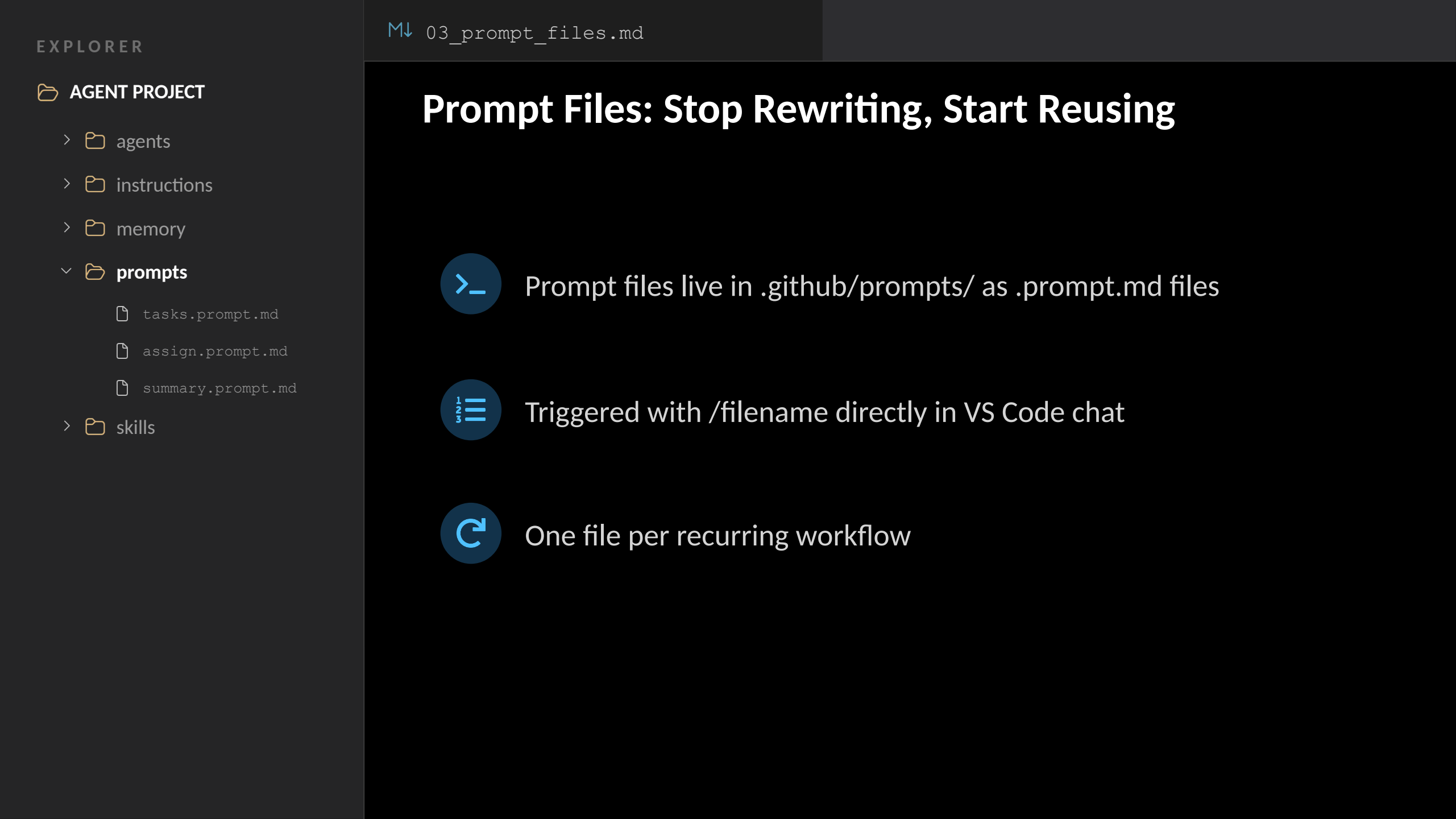


## Make it improve itself

Add an instruction inside the skill: if something goes wrong, propose an update to this file




*Every mistake becomes a permanent improvement, not something you fix once and forget*


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03_prompt_files.md
Prompt Files: Stop Rewriting, Start Reusing
Prompt files live in .github/prompts/ as .prompt.md files
Triggered with /filename directly in VS Code chat
One file per recurring workflow

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- skills

M↓ 03_prompt_files.md

# Prompt Files: Stop Rewriting, Start Reusing

## Example: 3 prompt files, a full development cycle

1

### Prioritize tasks

Connects to Azure DevOps and orders open tasks by priority

2

### Assign to the right agent

Picks the best fit agent, splits oversized tasks

3

### Summarize and update

Drafts a summary and updates Azure DevOps

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03_prompt_files.md

.github > prompts > 1.review-and-prioritize-iteration.prompt.md

```
---
description: Review all User Stories assigned to me in the current iteration and recommend a priority order with justification.
---

Review my assigned User Stories in **Iteration {{ITERATION_NUMBER}}** of the Azure DevOps project `P-AFI` for user `{{USER_EMAIL}}`.

For each US assigned to me:
1. Fetch its state, title, description, and acceptance criteria.
2. Identify dependencies on other US or tasks (blocked / blocking).
3. Assess implementation complexity (data model change, form/view, script, plugin, manual config).
4. Recommend a **priority order** with a brief justification for each item (why this one before the next).

Present the result as a ranked list with state, complexity estimate, and reason for the priority.
```

You, 6 seconds ago • Uncommitted changes

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03_prompt_files.md

.github > prompts > 2.agent-vs-manual-task-guide.prompt.md

You, last month | 1 author (You)

```
1  ---
2  description: For a given User Story, identify which tasks can be done by an agent and which require
   manual steps, with clear instructions for each.
3  ---
4
5  For the User Story **#{{US_ID}}** in Azure DevOps project `P-AFI-AquaPlus`:
6
7  1. Fetch the US description, acceptance criteria, and all child tasks.
8  2. For **each task**, classify it as:
9     - **Agent** — can be implemented automatically (e.g. Dataverse metadata, form/view via MCP,
   TypeScript web resource, ADO updates)
10    - **Manual** — requires human action in a UI (e.g. Power Platform Admin Center, maker portal
   sharing, test user login, Azure portal)
11 3. For **agent tasks**: describe exactly which agent/tool to use and what it will do.
12 4. For **manual tasks**: provide step-by-step instructions including which portal/screen to navigate
   to.
13
14 Present the result as a table per task, followed by the detailed manual instructions.
15
```

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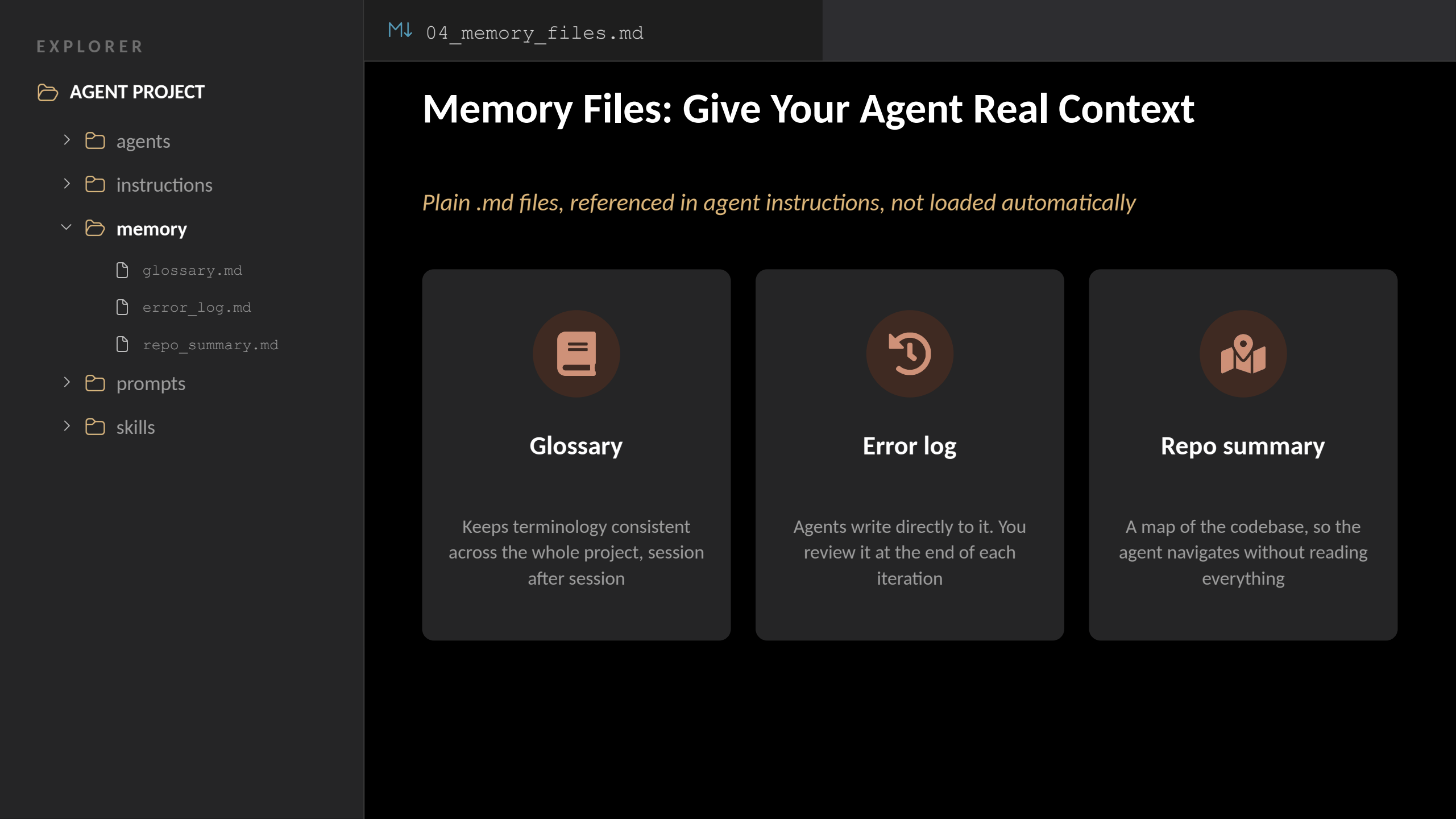
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03_prompt_files.md

.github > prompts > 3.add-completion-comment.prompt.md

You, last month | 1 author (You)

```
1  ---
2  description: Add a completion comment to one or more ADO tasks or User Stories after implementation,
   in English with German domain terms preserved.
3  ---
4
5  Add a completion comment to the following Azure DevOps work item(s) in project `P-AFI-AquaPlus`:
6
7  **Work item IDs:** {{WORK_ITEM_IDS}}
8
9  Write the comment in **English**, keeping German domain terms (e.g. Geschäftsfall, Fachspezialist,
   Anlage, Verfahren, Leser) in German with no translation.
10
11 The comment should include:
12 1. **What was implemented** — a short summary of the change made.
13 2. **How to test** — step-by-step instructions for the tester, referencing exact form names, fields,
   and expected behaviour.
14 3. **Notes** — any known limitations, dependencies on other tasks, or follow-up actions needed.
15
16 Format the comment as HTML (use `<strong>`, `<ul>/<li>`, `<p>` tags) because ADO renders HTML in
   comments.
17
18 After writing the comment, also update the task **State** to `Closed`.
19
```


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glossary.md
error_log.md
repo_summary.md
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04_memory_files.md
Memory Files: Give Your Agent Real Context
Plain .md files, referenced in agent instructions, not loaded automatically
Glossary
Keeps terminology consistent across the whole project, session after session
Error log
Agents write directly to it. You review it at the end of each iteration
Repo summary
A map of the codebase, so the agent navigates without reading everything

EXPLORER

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- skills

memory.md

```
// input — agents read

# Glossary

GWR — Grundwasserrecht.
Water rights managed at cantonal
level in Kanton Zürich.

SF — Schutzzone Förderbrunnen.
Protection zones around active
groundwater pump stations.

WR — Wasserrechtskonzession.
Concession to extract groundwater,
granted for a defined period.
```

free format

features.md

```
// output — agents write

| date   | agent    | error         | fix
|
|--------|----------|---------------|-----------
|
| 06-12  | Coord.   | token exp.    | env vars
|
| 06-13  | Transl.  | wrong: GWR    | glossary
|
| 06-14  | DataMod  | missing FK    | schema
|
| 06-15  | Plugins  | auth scope    | sp perms
|
```

needs structure

// free input · structured output

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04_memory_files.md

.github > memory > shared--lessons-learned.md > # Agent Lessons Learned > ## Log

Iñaki Albero Jorda, yesterday | 3 authors (You and others)

```
1  # Agent Lessons Learned
2
3  <!--
4  FILE RULES:
5  - Maximum 20 entries in the Log.
6  - When it reaches 20, consolidate by RELEVANCE before age: remove or merge entries that describe one-time flukes unlik
   g. environment setup issues, transient API errors). Never remove an entry that documents a recurring failure pattern o
   trap in the codebase or toolchain.
7  - Do not duplicate existing entries.
8  - Only NON-OBVIOUS errors and NON-TRIVIAL solutions.
9  -->
10
11 ## Format
12
13 - [agent] [YYYY-MM-DD] Situation → What failed → How it was resolved
14
15 ## Log
16
17 <!-- Entries are added here, most recent at the bottom -->
18
19 - [Dataverse Data Model Engineer] [2026-05-14] Creating Boolean columns on a German-language Dataverse org → `CreateBo
   MCP tool defaults to English labels (Yes/No) → Always pass `trueLabel: Ja` and `falseLabel: Nein` explicitly. Omitting
   delete-and-recreate since there is no UpdateColumn tool.
20 - [Dataverse UI Script Engineer] [2026-05-14] Running XrmDefinitelyTyped fails silently if `UserId` in `XrmDefinitely
   config` is stale → `removeUsername.ps1` only strips `Username=`, NOT `UserId=` → Always manually verify/update `UserId
   current developer's account before running `Run.ps1`.
21 - [Dataverse UI Script Engineer] [2026-06-30] Type-specific form routing for `ktzh_businesscase` looked correct by URL
   query param) but still loaded the default form → UCI can keep requested `formid` in URL while actually falling back to
   default form (often when target form is not available in app/role) → Use `formContext.ui.formSelector.getCurrentItem()
   the loaded-form source of truth and add a one-time session guard per record+target to prevent infinite onLoad navigati
22
```

# Glossar für Übersetzungsagenten

| German term | English term |
|---|---|
| Geschäftsfall | business case |
| Geschäftsfall-ID | business case ID |
| Geschäftsfalltyp | business case type |
| Gesuch | application |
| Gesuchsdokument | application document |
| Gesuchsformular | application form |
| Grundeigentümer | landowner |
| Grundwasserrechtsnummer | groundwater rights number |
| Katasternummer | cadastral number |
| Koordiniertes Verfahren | coordinated procedure |
| nicht-koordiniertes Verfahren | non-coordinated procedure |
| Konzessionär | concessionaire |
| konzessioniert | licensed |
| Parzelle | plot |
| Produkt | product |

# How Agents Consume Credits

*It's not one question. It's every token in between*

GitHub Copilot now bills by **token consumption**, effective since 1. June 2026

### Input tokens

Your message, files, context

### Output tokens

Everything the model generates

### Cached tokens

Cheaper, but they still count

Agent reads 10 files before answering





**10 files worth of input tokens**

*Bigger context means a more expensive bill on every single response*

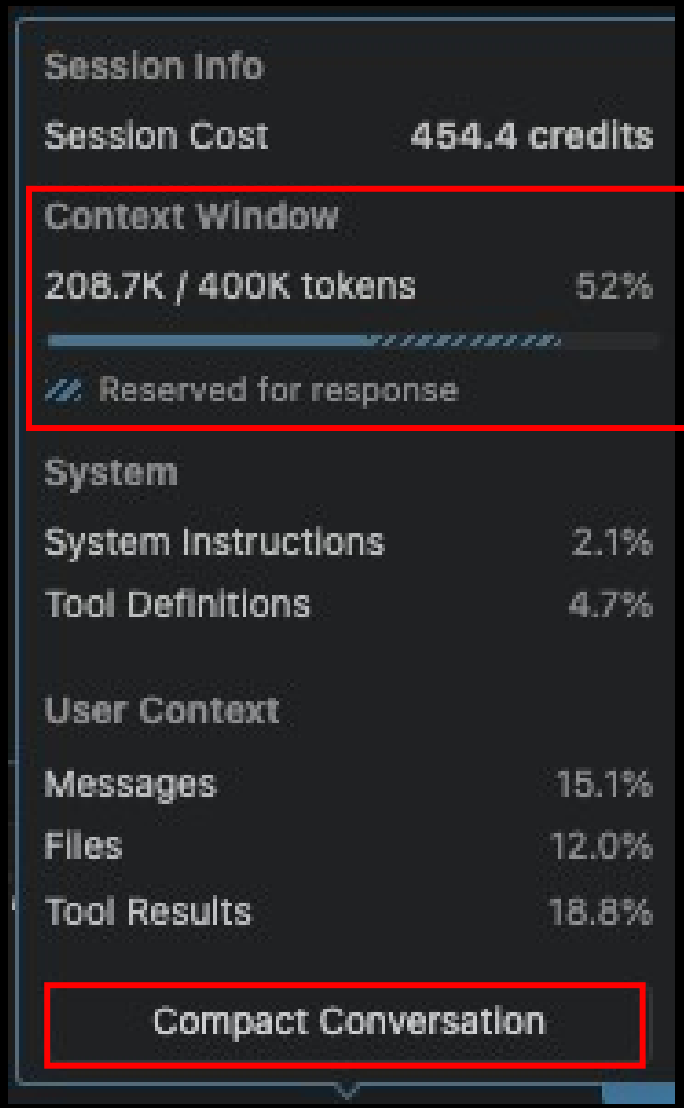
EXPLORER

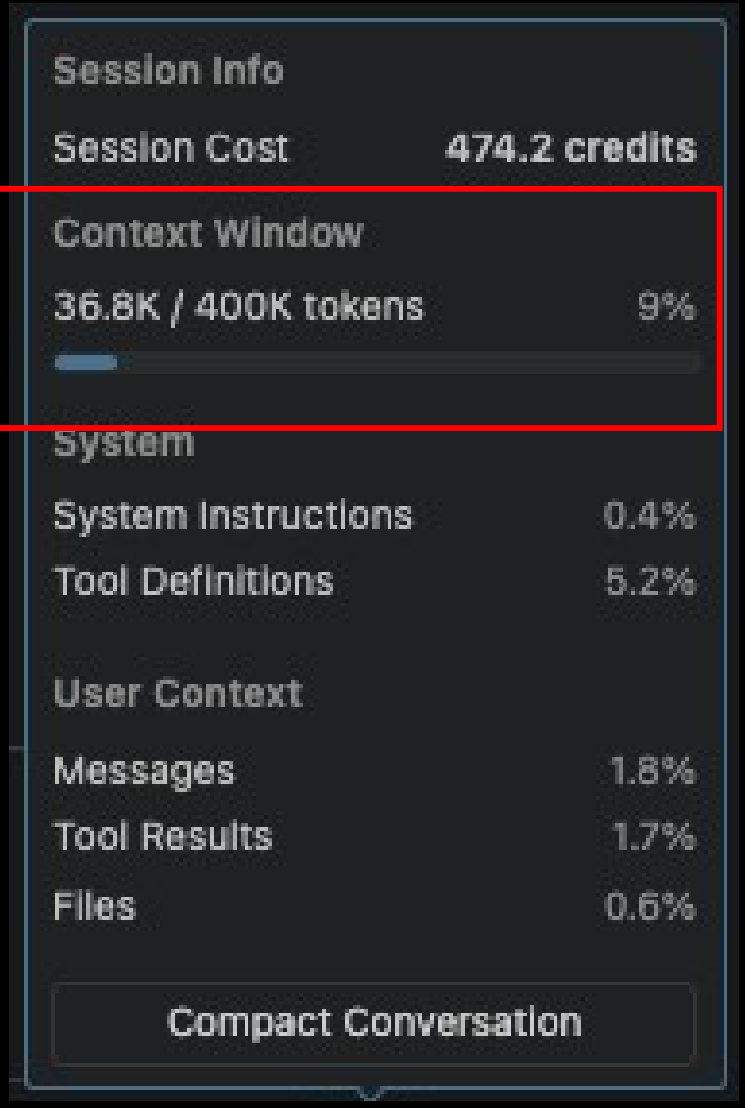
AGENT PROJECT

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05_credits.md

# How Agents Consume Credits



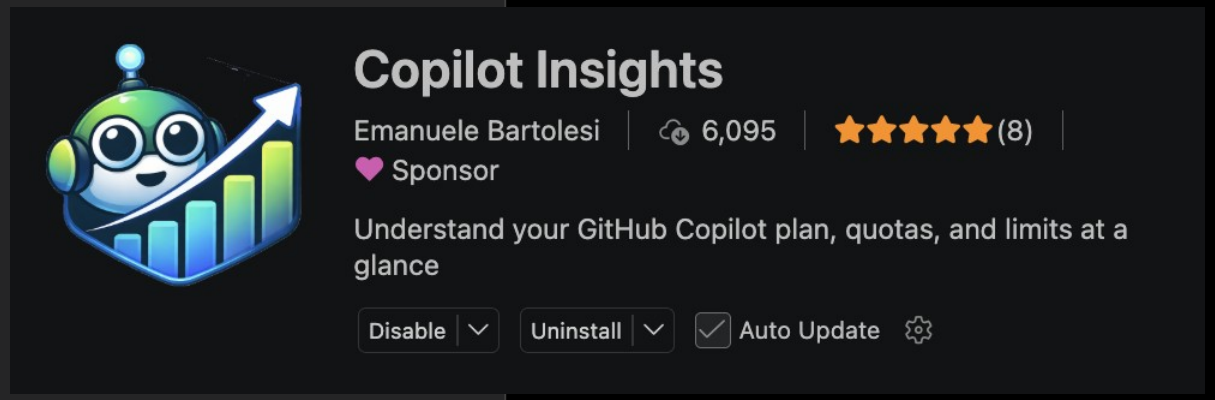
EXPLORER

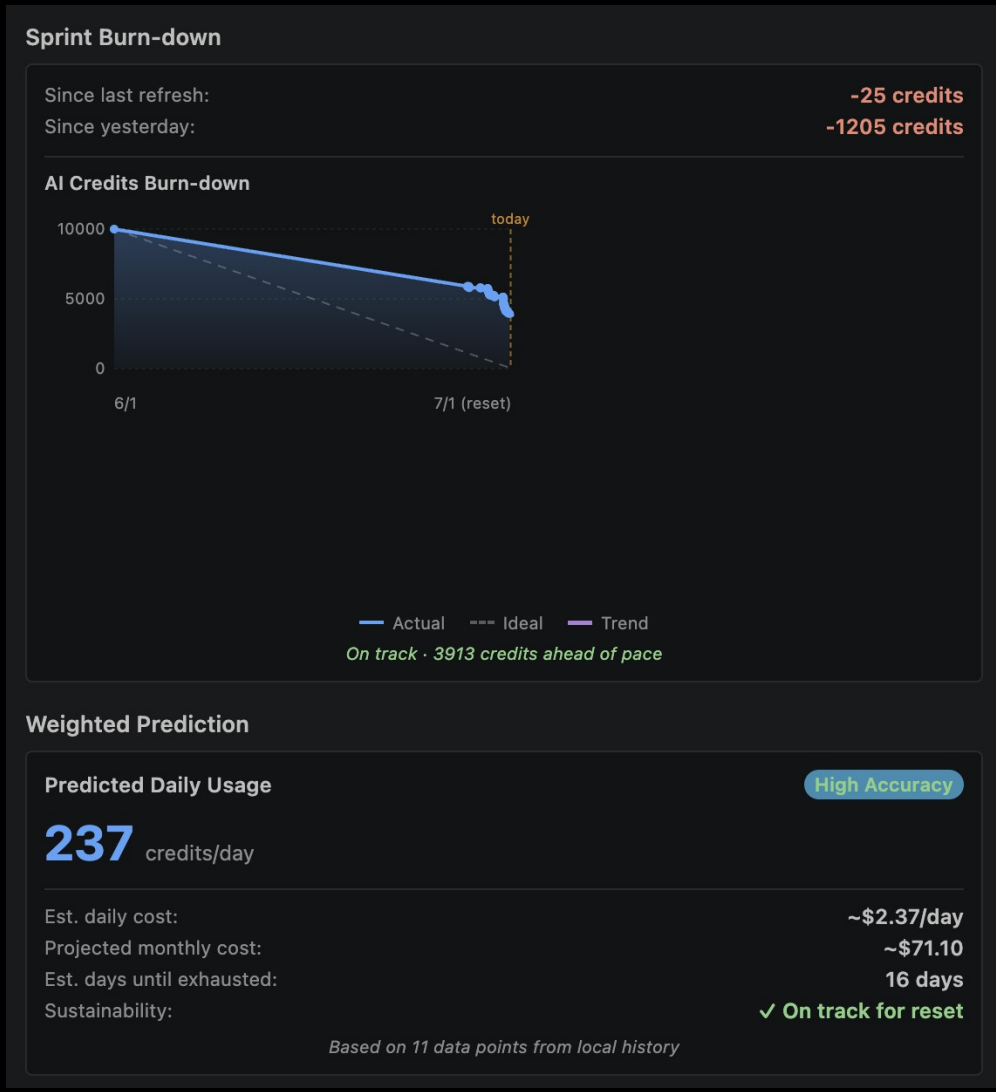
AGENT PROJECT

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M↓ 05_credits.md

# How Agents Consume Credits



# Choosing the Right Model

*Two axes: how well it reasons, and what it costs per interaction*

*Reasoning* ↑

| | |
|---|---|
| **Fast / cheap**<br>Routine edits, boilerplate, autocomplete | **Deep reasoning**<br>Hard debugging, architecture, real ambiguity |
| **General purpose**<br>Your default when you're not sure | **Higher multiplier**<br>Big refactors, use it on purpose |

*Cost* →

## How to decide in practice

- Most daily tasks don't need the most expensive model. That's a requirements decision, not a quality one
- A 3x multiplier costs three times what a 0.33x multiplier costs for the same interaction
- Auto now routes on the task itself, not just server load, and pays 10% less than picking the same model by hand

# Choosing the Right Model

## Claude Opus 4.6

Powerful | High cost

Credits Per 1M Tokens

| | |
|---|---|
| Input | 500 |
| Output | 2500 |
| Cache Read | 50 |
| Cache Write | 625 |

Configurable: Thinking Effort, Context Size

## GPT-5.3-Codex

Powerful | Medium cost

Credits Per 1M Tokens

| | |
|---|---|
| Input | 175 |
| Output | 1400 |
| Cache Read | 17 |
| Cache Write | 0 |

Configurable: Thinking Effort

## Claude Haiku 4.5

Lightweight | Low cost

Credits Per 1M Tokens

| | |
|---|---|
| Input | 100 |
| Output | 500 |
| Cache Read | 10 |
| Cache Write | 125 |

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06_choosing_model.md

# Choosing the Right Model

GPT-5.3-Codex • 24.9 credits

**Weighted Prediction**

**Predicted Daily Usage** — High Accuracy

**255** credits/day

| | |
|---|---|
| Est. daily cost: | ~$2.55/day |
| Projected monthly cost: | ~$76.50 |
| Est. days until exhausted: | 34 days |
| Sustainability: | ✓ On track for reset |

*Based on 8 data points from local history*

**Burn Rate Analysis**

1 AI credit = $0.01 USD

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06_choosing_model.md

# Choosing the Right Model

| Model | Release status | Category | Tier | Threshold (input tokens) | Input | Cached input | Output |
|---|---|---|---|---|---|---|---|
| GPT-5 mini | GA | Lightweight | Default | Not applicable | $0.25 | $0.025 | $2.00 |
| GPT-5.3-Codex | GA | Powerful | Default | Not applicable | $1.75 | $0.175 | $14.00 |
| GPT-5.4 | GA | Versatile | Default | ≤ 272K | $2.50 | $0.25 | $15.00 |
| GPT-5.4 | GA | Versatile | Long context | > 272K | $5.00 | $0.50 | $22.50 |
| GPT-5.4 mini | GA | Lightweight | Default | Not applicable | $0.75 | $0.075 | $4.50 |
| GPT-5.4 nano | GA | Lightweight | Default | Not applicable | $0.20 | $0.02 | $1.25 |
| GPT-5.5 | GA | Powerful | Default | ≤ 272K | $5.00 | $0.50 | $30.00 |
| GPT-5.5 | GA | Powerful | Long context | > 272K | $10.00 | $1.00 | $45.00 |

# The Model Line-up Moved Again

*The screenshots you just saw are already one generation behind*



## Claude Opus 5

In Copilot since 24 July 2026. Built for long, multi-step agent runs. Same list price as Opus 4.6: $5 in, $25 out per 1M tokens



## Claude Sonnet 5

The plot twist: cheaper than the model it replaces. $2 in, $10 out per 1M, versus $3 and $15 for Sonnet 4.6. Newer is not always pricier



## Retiring 1 Sep 2026

Opus 4.5, 4.6, 4.7, Sonnet 4.5, 4.6, Gemini 3.1 Pro and Raptor mini. If your team pinned a model in config, check it before September

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07_stay_in_control.md

# How to Keep it Under Control

*Work smarter, not wider*



## Repo summary

Agents go straight to what they need. No exploring, no scanning, no accumulating unnecessary context. One file read instead of ten



## Memory files

Terminology, error patterns, decisions: every piece of structured knowledge given upfront replaces an expensive exploration with a cheap, targeted read

***A focused agent is a cheap agent***

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M↓ 06c_local_models.md

# Local Models: Nothing Leaves the Machine

*Client-side local models in Copilot CLI: shipped 17 June 2026*



## Point the CLI at localhost

```
export COPILOT_PROVIDER_BASE_URL=http://localhost:11434
export COPILOT_MODEL=llama3.2
copilot        # no API key for a local Ollama instance
```



## Offline mode, for real isolation

COPILOT_OFFLINE=true stops the CLI contacting GitHub servers — for air-gapped work. The caveat worth saying out loud: it only isolates you if the provider is local too

***For a client who will not let code leave the building, this is the answer.***

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09_agent_plugins.md

# Agent Plugins 1.0: Build Once, Run Anywhere

*Generally available since 12 August 2026, on every Copilot plan*



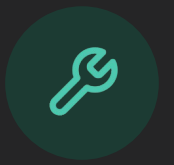
## One portable package

Skills and MCP servers bundle into one plugin. Instructions, skills and prompt files ship as a single unit instead of one layout per tool



## Every Copilot surface

VS Code, Copilot CLI, the SDK and the Copilot app. Install from the Awesome Copilot marketplace, a repo or a local path. No migration needed



## Governed, not loose

Business and Enterprise admins use enabledPlugins in managed-settings.json to install or block plugins across the org

# It's Not the First Attempt.
# It's the First of Many.

*The same philosophy you built into the system works for your daily practice with agents*



## The loop you built, apply it to yourself

The continuous improvement system for your skills and agents works the same way for your own practice. Build it once, run it always



## Don't get frustrated on the first try

When an agent doesn't work the first time, that is expected and part of the process. What we tell our teams: keep going



## If the system can fix it, you can always improve

A system ready to catch and correct its own errors gets better with every iteration, and so does the case for using agents

***Every iteration makes more sense than the last.***

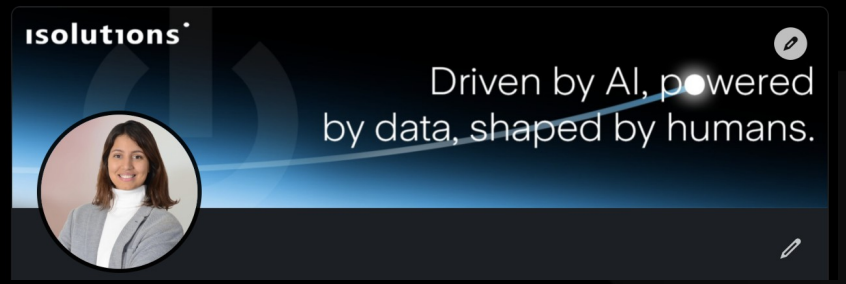
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# THANK YOU

Please give us a feedback on the session via the run.events App




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